# Conditions for Oscillation in the Wilson-Cowan Model

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***Abstract.*** *We investigate why the Neuromatch Academy (NMA) default parameters for the Wilson-Cowan model cannot produce oscillations under any external input. Starting from the Hopf bifurcation requirements on the Jacobian trace and determinant, we derive a structural necessary condition on the network parameters:* $w_{EI}w_{IE} > 16\tau_E w_{II}/(\tau_I a_E)$*. This single inequality, involving only coupling weights, time constants, and sigmoid gain, determines whether oscillation is possible regardless of external input or fixed-point tuning. The NMA parameters violate this condition (*$52 < 73.3$*), while the oscillatory parameters of Li et al. (2022) satisfy it with wide margin (*$520 \gg 32$*). We further develop a three-level verification framework — structural feasibility, window existence at realised fixed points, and actual operating point within the window — and validate it with a Monte Carlo study (*$N = 5000$*). The structural condition is confirmed as a strict necessary condition, with zero genuine theoretical violations across the entire framework in 5000 samples, while the full sufficient condition requires the unstable fixed point to be the system's unique equilibrium. Applying this framework to the Li et al. analysis, we show that excitatory self-feedback* $w_{EE}$ *acts as a tuning parameter with a finite effective range, clarifying the bounded oscillation window observed in their numerical bifurcation results.*

In computational neuroscience, the **Wilson-Cowan model** (Hugh R. Wilson & Jack D. Cowan) describes the dynamics of interactions between populations of very simple excitatory and inhibitory model neurons. It is a mean-field model that tracks the **proportion of active neurons** in each population over time.

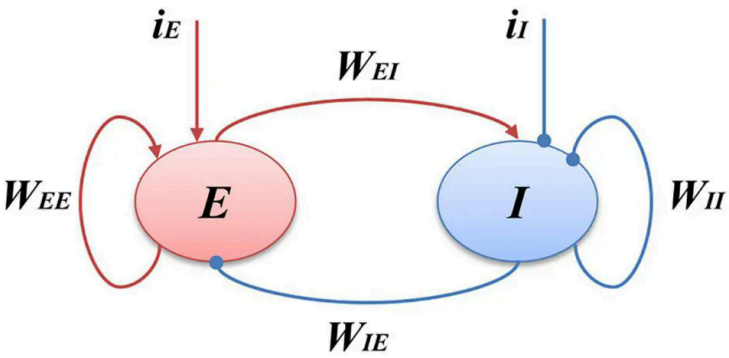


Figure 1: Schematic of the Wilson-Cowan model[1]

$$\tau_E \frac{dr_E}{dt} = -r_E + F_E(w_{EE} \cdot r_E - w_{EI} \cdot r_I + I_E^{\text{ext}}) \tag{1}$$

$$\tau_I \frac{dr_I}{dt} = -r_I + F_I(w_{IE} \cdot r_E - w_{II} \cdot r_I + I_I^{\text{ext}}) \tag{2}$$

where the activation function $F$ is a sigmoid:

$$F(x; a, \theta) = \frac{1}{1 + e^{-a(x-\theta)}} - \frac{1}{1 + e^{a\theta}} \tag{3}$$

[1] $E$ represents the excitatory populations, $I$ stands for the inhibitory populations, the red arrows indicate the excitatory projections and blue arrows show the inhibitory projections, $i_I$ is the external inputs of the $I$ populations, and $i_E$ is the external inputs of the $E$ populations. Li et al. (2022)

The correction term $-\frac{1}{1+e^{a\theta}}$ ensures $F(0) = 0$ (no input yields no activation).

> **Remark**: We use the simplified Wilson-Cowan equations without the refractory term $(1 - r_E)$, following the formulation in Neuromatch Academy.

## 0.1 Variables

- $r_E(t)$ — average firing rate (or proportion of active neurons) of the **excitatory** population at time $t$. Ranges from 0 (silent) to 1 (maximally active).
- $r_I(t)$ — average firing rate of the **inhibitory** population at time $t$.

## 0.2 Parameters

### 0.2.1 Time Constants

- $\tau_E$ — time constant of the excitatory population.
- $\tau_I$ — time constant of the inhibitory population.

### 0.2.2 Sigmoid Parameters

- $a_E$, $a_I$ — gain (steepness) of the activation function for E and I populations. Higher gain means a sharper threshold-like response.
- $\theta_E$, $\theta_I$ — threshold of the activation function. The input level at which the population reaches half-maximal activation.

### 0.2.3 Connection Weights

- $w_{EE}$ — excitatory $\rightarrow$ excitatory coupling strength (recurrent excitation).
- $w_{EI}$ — inhibitory $\rightarrow$ excitatory coupling strength (how much I suppresses E).
- $w_{IE}$ — excitatory $\rightarrow$ inhibitory coupling strength (how much E drives I).
- $w_{II}$ — inhibitory $\rightarrow$ inhibitory coupling strength (recurrent inhibition).

### 0.2.4 External Inputs

- $I_E^{\text{ext}}$ — external input current to the excitatory population (primary control parameter for bifurcation analysis).
- $I_I^{\text{ext}}$ — external input current to the inhibitory population (typically fixed at 0).

### 0.2.5 Parameter Values

| Parameter | Value | Description |
|---|---|---|
| $\tau_E$ | 1.0 ms | Average synaptic time constant for excitatory population |
| $\tau_I$ | 2.0 ms | Average synaptic time constant for inhibitory population |
| $a_E$ | 1.2 | Excitatory gain |
| $a_I$ | 1.0 | Inhibitory gain |
| $\theta_E$ | 2.8 | Excitatory threshold |
| $\theta_I$ | 4.0 | Inhibitory threshold |
| $w_{EE}$ | 9.0 | Strength of self-excitatory feedback |
| $w_{EI}$ | 4.0 | Strength of connections between E–I |
| $w_{IE}$ | 13.0 | Strength of connections between I–E |
| $w_{II}$ | 11.0 | Strength of self-inhibitory feedback |
| $I_E^{\text{ext}}$ | 0.0 | External input to excitatory population (sweep this) |
| $I_I^{\text{ext}}$ | 0.0 | External input to inhibitory population |

Table 1: Default parameter values from the Neuromatch Academy computational neuroscience curriculum.

**Remark**: The parameters play an important role in the system, please read the *Periodic solution (Oscillations)* part in details.

# 1 Nullcline Analysis

## 1.1 E-nullcline

Let $\dot{r}_E = 0$, then

$$\frac{1}{\tau_E}(-r_E + F_E(w_{EE} \cdot r_E - w_{EI} \cdot r_I + I_E^{\text{ext}})) = 0 \tag{4}$$

$$-r_E + F_E(w_{EE} \cdot r_E - w_{EI} \cdot r_I + I_E^{\text{ext}}) = 0 \tag{5}$$

$$F_E(w_{EE} \cdot r_E - w_{EI} \cdot r_I + I_E^{\text{ext}}) = r_E \tag{6}$$

Then we use the inverse of $F$ (namely the logit function):

$$F^{-1}(y; a, \theta) = \theta + \frac{1}{a} \ln \frac{y + c}{1 - y - c}, \quad c = \frac{1}{1 + e^{a\theta}} \tag{7}$$

Applying $F_E^{-1}$ to both sides:

$$F_E^{-1}(r_E) = w_{EE} \cdot r_E - w_{EI} \cdot r_I + I_E^{\text{ext}} \tag{8}$$

Solving for $r_I$:

$$r_I = \frac{1}{w_{EI}}(w_{EE} \cdot r_E - F_E^{-1}(r_E) + I_E^{\text{ext}}) \tag{9}$$

### 1.2 I-nullcline

Analogously, let $\dot{r}_I = 0$, we have

$$r_I = F_I(w_{IE} \cdot r_E - w_{II} \cdot r_I + I_I^{\text{ext}}) \tag{10}$$

$$r_E = \frac{1}{w_{IE}}(w_{II} \cdot r_I + F_I^{-1}(r_I) - I_I^{\text{ext}}) \tag{11}$$

### 1.3 Phase plane

After substituting the parameters, we obtain the figure

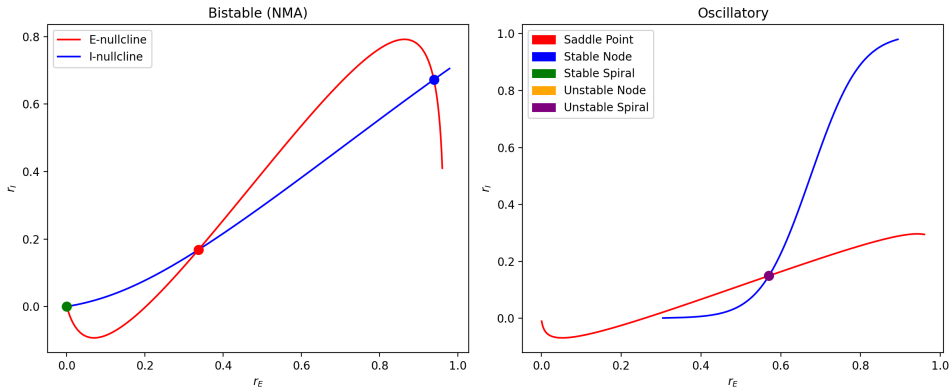


Figure 2: Phase plane analysis

Inserting point $(0.1, 0.5)$, it's easy to check the arrow direction. We omit here.

**Remark:** The domains are $r_E \in (0, 0.9664)$ and $r_I \in (0, 0.9820)$, as the logarithm requires its argument to be positive.

## 2 Qualitative Analysis

Note that in the previous section, we obtain the phase plane with 3 equilibrium points. Now we use the Jacobian matrix to determine its state. Recall the Jacobian matrix:

$$J = \begin{pmatrix} \frac{\partial \dot{r}_E}{\partial r_E} & \frac{\partial \dot{r}_E}{\partial r_I} \\ \frac{\partial \dot{r}_I}{\partial r_E} & \frac{\partial \dot{r}_I}{\partial r_I} \end{pmatrix} \tag{12}$$

We solve the partial derivative and obtain

$$J = \begin{pmatrix} \frac{1}{\tau_E}(-1 + w_{EE} \cdot F_E') & \frac{1}{\tau_E}(-w_{EI} \cdot F_E') \\ \frac{1}{\tau_I}(w_{IE} \cdot F_I') & \frac{1}{\tau_I}(-1 - w_{II} \cdot F_I') \end{pmatrix} \tag{13}$$

in which

$$F'(x) = a \cdot (F(x) + c)(1 - F(x) - c), \quad c = \frac{1}{1 + e^{a\theta}} \tag{14}$$

Note that $F'$ is evaluated at the sigmoid input, not at $r_E$ or $r_I$ directly:

$$u_E = w_{EE} r_E - w_{EI} r_I + I_E^{\text{ext}}, \quad u_I = w_{IE} r_E - w_{II} r_I + I_I^{\text{ext}} \tag{15}$$

We classify each fixed point using the trace $\tau = \text{tr}(J)$, determinant $\Delta = \det(J)$, and discriminant $\tau^2 - 4\Delta$ of the Jacobian:

- $\Delta < 0 \Rightarrow$ saddle point
- $\Delta > 0, \tau < 0 \Rightarrow$ stable (node if $\tau^2 - 4\Delta > 0$, spiral if $< 0$)
- $\Delta > 0, \tau > 0 \Rightarrow$ unstable

For our 3 fixed points:

| **Fixed Point** | $\tau$ | $\Delta$ | $\tau^2 - 4\Delta$ | **Classification** |
|---|---|---|---|---|
| $(0, 0)$ | $-1.247$ | $0.406$ | $-0.069$ | Stable Spiral |
| $(\approx 0.34, 0.17)$ | — | $-0.941$ | — | Saddle Point |
| $(\approx 0.94, 0.68)$ | $-2.403$ | $1.381$ | $0.250$ | Stable Node |

Table 2: Fixed point classification for default parameters ($I_E^{\text{ext}} = 0$).

The system with default parameters ($I_E^{\text{ext}} = 0$) is therefore **bistable**. (In a dynamical system, **bistability** means the system has two stable equilibrium states a bistable structure can be resting in either of two states.)

# 3 Periodic Solutions (Oscillations)

Recall the **Poincaré-Bendixson theorem** states that for a two-dimensional system, for which we know that the solutions will stay in **a bounded area** after some finite time and there is **no fixed point**, there must exist a periodic solution to which the trajectories converge.

We sweep the $I_E^{\text{ext}}$ until we obtain only one fixed point which is unstable.

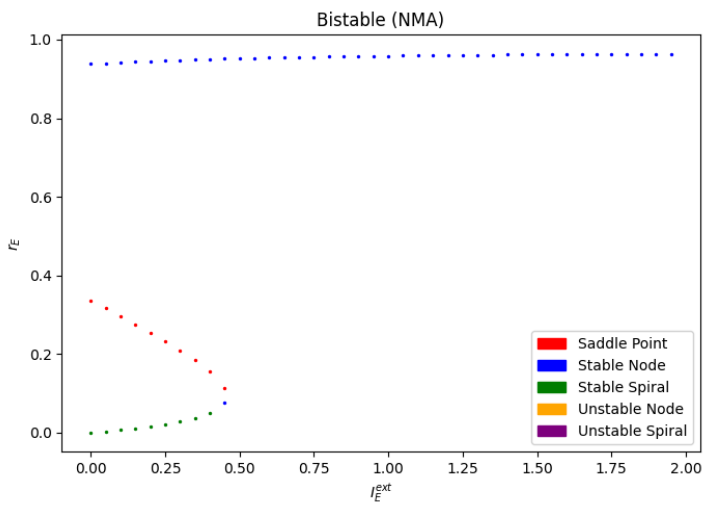


Figure 3: Stable fixed point output after sweeping $I_E^{\text{ext}}$

Note that after sweeping $I_E^{\text{ext}}$, we only get the stable 1 fixed point, why?

The answer lies in the network's structural parameters.

Comparing with the parameters used in Li et al. (2022) (Table 1, based on Wilson and Cowan (1972) and Jadi and Sejnowski (2014)) for gamma oscillation research:

| **Parameter** | **NMA (bistability)** | **Table 1 (oscillation)** |
|---|---|---|
| $w_{EE}$ | 9 | 16 |
| $w_{EI}$ | 4 | 26 |
| $w_{IE}$ | 13 | 20 |
| $w_{II}$ | 11 | 1 |
| $\tau_E$ (ms) | 1 | 20 |
| $\tau_I$ (ms) | 2 | 10 |
| $\theta_E, \theta_I$ | 2.8, 4.0 | 5, 20 |
| $I_E^{\text{ext}}, I_I^{\text{ext}}$ | 0, 0 | 2, 7 |

Table 3: Parameter comparison: bistable vs. oscillatory regime.

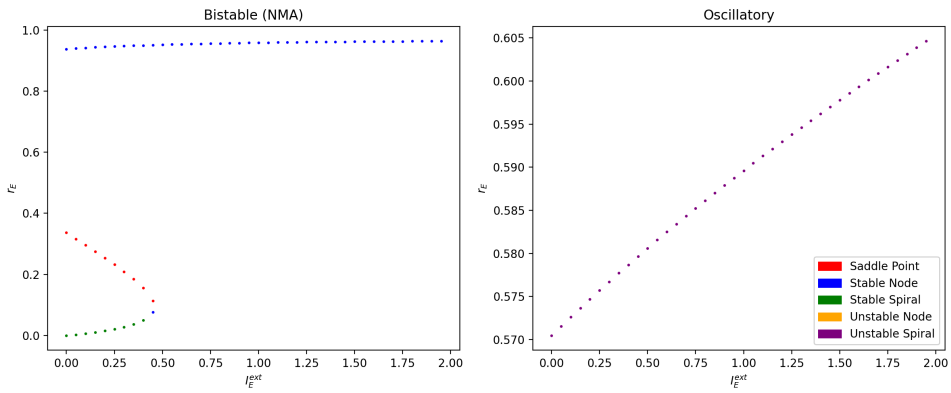


Figure 4: Output comparison between NMA (bistable) and Li et al. (oscillatory) parameter regimes.

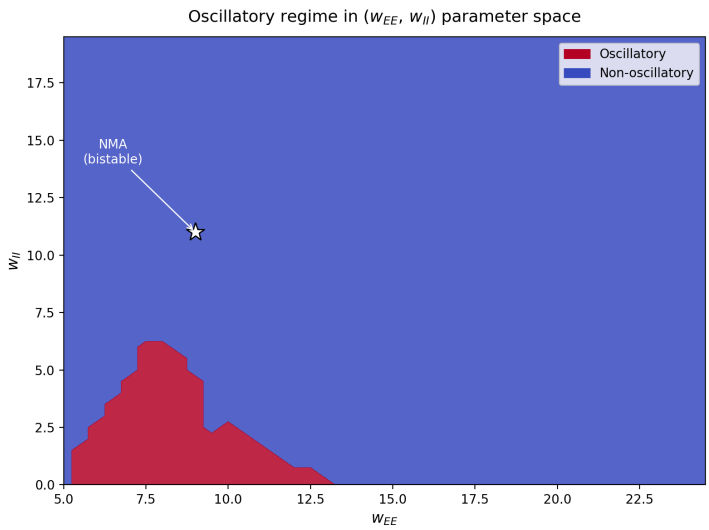


Figure 5: Oscillatory regime in (Wee, WI) parameter space for NMA parameters

## 3.1 Conditions for Oscillation

A Hopf bifurcation requires a fixed point where the Jacobian has both positive trace (instability) and positive determinant (complex eigenvalues):

$$\tau = \frac{1}{\tau_E}(-1 + w_{EE}F'_E) + \frac{1}{\tau_I}(-1 - w_{II}F'_I) \tag{16}$$

$$\Delta = \frac{1}{\tau_E\tau_I}[1 + w_{II}F'_I - w_{EE}F'_E + (w_{EI}w_{IE} - w_{EE}w_{II})F'_E F'_I] \tag{17}$$

- $\tau > 0$: In order to let $\frac{1}{\tau_E}(-1 + w_{EE}F'_E) + \frac{1}{\tau_I}(-1 - w_{II}F'_I) > 0$, we need

$$w_{EE}F'_E > 1 + \frac{\tau_E}{\tau_I}(1 + w_{II}F'_I) \tag{18}$$

- $\Delta > 0$: In order to let

$$\frac{1}{\tau_E\tau_I}[1 + w_{II}F'_I - w_{EE}F'_E + (w_{EI}w_{IE} - w_{EE}w_{II})F'_E F'_I] > 0 \tag{19}$$

  we require

$$1 + w_{II}F'_I + w_{EI}w_{IE}F'_E F'_I > w_{EE}F'_E + w_{EE}w_{II}F'_E F'_I \tag{20}$$

$$1 + w_{II}F'_I + w_{EI}w_{IE}F'_E F'_I > w_{EE}F'_E(1 + w_{II}F'_I) \tag{21}$$

  Rearranging as an upper bound on $w_{EE}F'_E$:

$$w_{EE}F'_E < \frac{1 + w_{II}F'_I + w_{EI}w_{IE}F'_E F'_I}{1 + w_{II}F'_I} \tag{22}$$

Combining Equation 18 and Equation 22, a feasibility window for $w_{EE}F'_E$ exists only when the lower bound is less than the upper bound:

$$1 + \frac{\tau_E}{\tau_I}(1 + w_{II}F'_I) < \frac{1 + w_{II}F'_I + w_{EI}w_{IE}F'_E F'_I}{1 + w_{II}F'_I} \tag{23}$$

Multiplying both sides by $(1 + w_{II}F'_I)$ and simplifying:

$$w_{EI}w_{IE} > \frac{\tau_E}{\tau_I} \cdot \frac{(1 + w_{II}F'_I)^2}{F'_E F'_I} \tag{24}$$

Since $F'_E$ and $F'_I$ are not free parameters but determined by the fixed point, we minimise the right-hand side to obtain the weakest necessary condition.

**Maximising $F'_E$.** $F'_E$ appears only in the denominator, so we need its maximum. From Equation 14:

$$F'(x) = a \cdot (F(x) + c)(1 - F(x) - c) \tag{25}$$

This is maximised when $F(x) + c = 1/2$, i.e. at the sigmoid midpoint $u_E = \theta_E$. Substituting:

$$F'^{\max}_E = a_E \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{a_E}{4} \tag{26}$$

**Minimising over $F'_I$.** Let $\varphi = F'_I > 0$. The right-hand side of Equation 24 becomes:

$$R(\varphi) = \frac{\tau_E}{\tau_I} \cdot \frac{(1 + w_{II}\varphi)^2}{(a_E/4) \cdot \varphi} = \frac{4\tau_E}{\tau_I a_E} \cdot \frac{1 + 2w_{II}\varphi + w_{II}^2\varphi^2}{\varphi} \tag{27}$$

Expanding:

$$R(\varphi) = \frac{4\tau_E}{\tau_I a_E}\left(\frac{1}{\varphi} + 2w_{II} + w_{II}^2\varphi\right) \tag{28}$$

Differentiating and setting to zero:

$$R'(\varphi) = \frac{4\tau_E}{\tau_I a_E}\left(-\frac{1}{\varphi^2} + w_{II}^2\right) = 0 \tag{29}$$

$$\varphi^2 = \frac{1}{w_{II}^2} \quad \Longrightarrow \quad \varphi^* = \frac{1}{w_{II}} \tag{30}$$

To confirm this is a minimum:

$$R''(\varphi) = \frac{4\tau_E}{\tau_I a_E} \cdot \frac{2}{\varphi^3} > 0 \quad \text{for all } \varphi > 0 \tag{31}$$

Substituting $F_E' = a_E/4$ and $F_I' = 1/w_{II}$:

$$R(\varphi^*) = \frac{4\tau_E}{\tau_I a_E}(w_{II} + 2w_{II} + w_{II}) = \frac{4\tau_E}{\tau_I a_E} \cdot 4w_{II} = \frac{16\tau_E w_{II}}{\tau_I a_E} \tag{32}$$

This yields the structural necessary condition:

$$w_{EI}w_{IE} > \frac{16\tau_E w_{II}}{\tau_I a_E} \tag{33}$$

Due to the $F_E'$ and $F_I'$ are jointly determined by the fixed-point coordinates, they cannot in general simultaneously attain their optimal values. This makes Equation 33 a potentially loose bound: a gap quantified by the three-level framework in the section 3.1.1.

This is a purely structural necessary condition: if it fails, no fixed point can undergo Hopf bifurcation regardless of where it sits on the sigmoid. We verify against both parameter sets:

- **Li et al.**: $\frac{16\times20\times1}{10\times1} = 32$, and $w_{EI}w_{IE} = 520 \gg 32$. ✓
- **NMA**: $\frac{16\times1\times11}{2\times1.2} = 73.3$, and $w_{EI}w_{IE} = 52 < 73.3$. ×

The role of each parameter in Equation 24 is:

- $w_{EI}w_{IE}$ (left-hand side): the cross-population coupling is the sole resource available to meet the cost. It represents the strength of the $E \to I \to E$ feedback loop.
- $\tau_E/\tau_I$: the time-constant ratio sets the baseline cost. When $\tau_E > \tau_I$ (inhibition recruited faster, the "weakly modulated" regime of Keeley et al. (2019)), the cost is higher.
- $(1 + w_{II}F_I')^2$: inhibitory self-feedback raises the cost quadratically.
- $F_E'$ (denominator only): the excitatory population must operate near its sensitive region ($u_E \approx \theta_E$) to maximise $F_E'$ and keep the cost manageable.
- $F_I'$ (both numerator and denominator): increasing $F_I'$ lowers the cost through the denominator but raises it through $(1 + w_{II}F_I')^2$ in the numerator. The optimal balance is at $F_I' = 1/w_{II}$.

> From Keeley et al. (2019): "…weakly modulated network gamma is found when the firing rate of the inhibitory population is recruited faster than that of the excitatory population, and strongly synchronized gamma oscillations are seen for the reverse. In both frameworks, network oscillations arise when a steady state loses stability via Hopf bifurcation…"

An immediate corollary concerns the cross-term $w_{EI}w_{IE}$ vs $w_{EE}w_{II}$. When $\tau_E > \tau_I$, assume $w_{EI}w_{IE} \leq w_{EE}w_{II}$. Then Equation 21 requires $1 + w_{II}F_I' > w_{EE}F_E'$. Combined with Equation 18:

$$1 + w_{II}F_I' > 1 + \frac{\tau_E}{\tau_I}(1 + w_{II}F_I') \tag{34}$$

Dividing by $(1 + w_{II}F_I') > 0$ gives $\tau_E < \tau_I$, a contradiction. Therefore, in the weakly modulated regime, $w_{EI}w_{IE} > w_{EE}w_{II}$ is strictly necessary.

### 3.1.1 From necessity to sufficiency

The structural condition assumes both sigmoid derivatives simultaneously take their most favourable values. In practice, the fixed-point position determines the actual $F_E'$ and $F_I'$, and the full master inequality (Equation 24) must be satisfied at a realised fixed point. This introduces two additional requirements:

- **Trace bound** ($\tau > 0$): $w_{EE}F_E'$ must exceed the threshold in Equation 18. This requires $w_{EE}$ to be large enough to place $u_E$ near $\theta_E$, keeping $F_E'$ near its maximum. But if $w_{EE}$ is too large, saturation collapses $F_E'$ and the trace condition fails from the other direction. The trace thus imposes both a lower and an upper bound on $w_{EE}$ through the behaviour of $F_E'$.
- **Determinant upper bound** ($\Delta > 0$): independently, $w_{EE}F_E'$ must not exceed the algebraic ceiling in Equation 22. This bound can be violated even when $F_E'$ is still large enough for $\tau > 0$ — the determinant turns negative while the trace remains positive, producing a Saddle.

Together with the structural condition, these requirements fully characterise the oscillatory regime at three levels:

1. **Structural** (Equation 33): can the network support oscillation at all? A necessary condition on coupling weights and time constants alone.
2. **Window exists** (Equation 24 at realised fixed point): does the feasibility window $L < U$ remain open at the actual sigmoid slopes? Necessary but not sufficient.
3. **Fixed point in window** ($L < w_{EE}F_E' < U$): does the actual $w_{EE}F_E'$ fall between the trace lower bound and the determinant upper bound? This guarantees the fixed point is an unstable spiral (local Hopf condition). **A limit cycle additionally requires that no other stable attractor captures the trajectories**, which is satisfied when this is the system's only fixed point[2].

[2] We discussed about it in section 3.2.3.

[3] To verify it, use our condition test module.

## 3.2 Verification[3]

### 3.2.1 Structural necessary condition

We first check Equation 33 against both parameter sets:

| | **NMA** | **Li et al.** |
|---|---|---|
| $w_{EI}w_{IE}$ | 52 | 520 |
| $\frac{16\tau_E w_{II}}{\tau_I a_E}$ | 73.3 | 32 |
| **Result** | $52 < 73.3$ × | $520 \gg 32$ ✓ |

Table 4: Structural necessary condition (Equation 33). NMA fails even under the most favourable sigmoid slopes, ruling out Hopf bifurcation for any $w_{EE}$ or $I_E^{\text{ext}}$.

For Li et al., the condition is satisfied with wide margin, and the system indeed oscillates at the default parameters ($u_E = 5.24 \approx \theta_E = 5.0$, confirming that $w_{EE}$ is tuned to keep $F'_E$ near its maximum).

### 3.2.2 Fixed points

We verify across the $(w_{EE}, w_{II})$ parameter space using NMA cross-coupling values ($w_{EI} = 4$, $w_{IE} = 13$):

| $w_{EE}$ | $w_{II}$ | $I_E^{\text{ext}}$ | **Structural** | **Window** | **In window** | **Observed behaviour** |
|---|---|---|---|---|---|---|
| 11 | 0 | 0.50 | ✓ | ✓ | ✓ | **Unstable Spiral** (oscillation) |
| 11 | 1 | 0.50 | ✓ | ✓ | ✓ | **Unstable Node** (oscillation) |
| 11 | 5 | 0.00 | ✓ | × | × | Stable Spiral + Saddle + Stable Node |
| 14 | 2 | 0.00 | ✓ | ✓ | × | Stable Spiral + Saddle |
| 20 | 0 | 0.00 | ✓ | × | × | Stable Node + Saddle |
| 9 | 11 | 0.00 | × | × | × | Stable Spiral + Saddle + Stable Node |

Table 5: Three-level verification of oscillation conditions.[4]

Each failure mode corresponds to a different level:

- $(w_{EE} = 9, w_{II} = 11)$ (NMA defaults): fails at level 1. The structural condition gives $52 < 73.3$, ruling out oscillation regardless of $w_{EE}$ tuning or external input.
- $(w_{EE} = 11, w_{II} = 5)$: passes level 1 ($52 > 33.3$), but fails at level 2. Increasing $w_{II}$ shifts the fixed points so that the actual $F'_E$ and $F'_I$ deviate from their optimal values, collapsing the feasibility window.
- $(w_{EE} = 20, w_{II} = 0)$: passes level 1 ($52 > 0$), but fails at level 2. Saturation pushes $F'_E \approx 0$ at both fixed points, inflating the right-hand side of Equation 24 far beyond $w_{EI}w_{IE} = 52$.

[4] **Structural**: Equation 33 satisfied.
**Window**: feasibility window $L < U$ exists at the realised fixed point (Equation 24).
**In window**: actual $w_{EE}F'_E$ falls between trace lower bound and determinant upper bound.
Oscillation requires all three.

- $(w_{EE} = 14, w_{II} = 2)$: passes level 1 ($52 > 13.3$) and level 2 — the window exists. But fails at level 3: the saddle achieves $\tau > 0$ ($w_{EE}F'_E$ exceeds the lower bound), yet $w_{EE}F'_E$ also exceeds the algebraic ceiling from Equation 22, pushing $\Delta < 0$. The actual operating point overshoots the window from above.
- $(w_{EE} = 11, w_{II} = 0)$ and $(11, 1)$: all three levels satisfied. $w_{EE}$ places $u_E$ near $\theta_E$, keeping $F'_E$ near-maximal, and the actual $w_{EE}F'_E$ falls within the feasibility window.

### 3.2.3 Monte Carlo validation

To test the framework beyond hand-picked parameters, we sampled 5000 random parameter combinations uniformly across wide ranges ($w_{EE}, w_{EI}, w_{IE} \in [1, 30]$, $w_{II} \in [0, 20]$, $\tau_E, \tau_I \in [0.5, 30]$, $a_E, a_I \in [0.5, 3]$, $\theta_E \in [1, 10]$, $\theta_I \in [1, 20]$, $I_E^{\text{ext}}, I_I^{\text{ext}} \in [-2, 5]$) and compared the analytical predictions against numerical integration with oscillation detection.

| Category | Count | Oscillating | Rate |
|---|---|---|---|
| Level 1 fail | 1698 | 0 | 0.00% |
| Level 1 pass, Level 2 fail | 2613 | 1 | 0.04% |
| Level 1–2 pass, Level 3 fail | 351 | 15 | 4.27% |
| Level 3 pass | 338 | 142 | 42.01% |

Table 6: Monte Carlo validation ($N = 5000$)[5].

The structural necessary condition (Level 1) is fully confirmed: none of the 1698 samples that failed it produced oscillation. The single Level 2 anomaly (1 out of 2613) has parameters $w_{EE} = 18.8$, $w_{II} = 13.6$, $\tau_E = 5.1$, $\tau_I = 14.4$; the system has a single stable fixed point (no unstable point), making a true limit cycle impossible. This is a false positive from the oscillation detector.

Similarly, all 15 middle-zone oscillations and the single Level 2 anomaly involve systems with no unstable fixed point (U0/S1/Sad0 or U0/S2/Sad1). These are false positives from the oscillation detector — slowly decaying spirals near the Hopf boundary — not theoretical violations.

The Level 3 confirmation rate of 42% reveals an important refinement. Level 3 guarantees that a fixed point is locally unstable (Hopf condition), but not that a limit cycle forms. In multi-fixed-point systems, an unstable spiral can coexist with stable attractors that capture trajectories before a limit cycle develops. To verify this interpretation, we partition the Level 3 passes by fixed-point count:

| Fixed-point structure | Count | Oscillating | Rate |
|---|---|---|---|
| 1 fixed point (unstable) | 145 | 130 | 89.7% |
| 3 fixed points | 162 | 10 | 6.2% |
| Other | 31 | 2 | 6.5% |

Table 7: Level 3 passes partitioned by fixed-point count.[6]

The 10 oscillating three-fixed-point cases all share the classification U1/S1/Sad1 (one unstable spiral, one stable attractor, one saddle). No U2/S0/Sad1 configuration — in which a

[5] Oscillation detected via numerical integration (RK4) with periodicity analysis after transient removal.

[6] Systems with a unique unstable fixed point oscillate at $\approx 90\%$; the gap from 100% is attributable to numerical limitations (fixed-point detection and oscillation sensitivity). Multi-fixed-point systems rarely oscillate despite passing Level 3, confirming that competing stable attractors suppress limit cycle formation.

global limit cycle could enclose all three fixed points — was observed. This indicates that the observed oscillations are basin-restricted[7]: trajectories initiated near the unstable spiral orbit within its basin before eventually being captured by the competing stable attractor, producing transient oscillatory behaviour that the detector registers as periodic.

**Across all 5000 samples, zero genuine violations of the analytical framework were observed.**

The complete sufficient condition for oscillation in the Wilson-Cowan model is therefore:

*Level 3 satisfied at the system's **unique** fixed point.*

When a single fixed point exists and is an unstable spiral, the system is bounded ($r_E, r_I \in [0,1]$)[8] and contains no other attractor, so the Poincaré-Bendixson theorem guarantees a limit cycle.

## 3.3 Further discussion

Our analytical conditions are consistent with the numerical bifurcation results of Li et al. (2022), who investigated the role of self-feedback loops in regulating gamma oscillations in the Wilson-Cowan model.

Their one-parameter bifurcation with respect to $w_{II}$ (their Figure 3A) shows that increasing inhibitory self-feedback suppresses oscillation. This follows directly from the structural condition (Equation 33): the right-hand side $\frac{16\tau_E w_{II}}{\tau_I a_E}$ grows linearly with $w_{II}$, so for fixed cross-coupling $w_{EI}w_{IE}$, sufficiently large $w_{II}$ will always violate the inequality. In the full master inequality (Equation 24), $w_{II}$ enters quadratically through $(1 + w_{II}F_I')^2$, making its suppressive effect even stronger at realised fixed points.

For excitatory self-feedback, their one-parameter bifurcation with respect to $w_{EE}$ (their Figure 6A) reveals that oscillation exists only within a limited window ($13.57 < w_{EE} < 35$). Their study concludes that "the excitatory self-feedback loop promotes the generation of gamma oscillations". However, this statement captures only the lower boundary of the window.

Their transfer function analysis correctly identifies the qualitative trend: increasing $w_{EE}$ drives the excitatory population toward instability. Quantitatively, however, the closed-loop gain formula (their Eq. 13) requires $w_{EE}F_E' < 1$, while the Hopf trace condition demands $w_{EE}F_E' > 1$. The two-dimensional Jacobian analysis developed here covers the oscillatory regime directly without this restriction.

Since $w_{EE}$ does not appear in the structural condition (Equation 33), its role is not to determine *whether* the network can support oscillation — that is governed by $w_{EI}w_{IE}$, $w_{II}$, $\tau_E/\tau_I$, and $a_E$. Rather, $w_{EE}$ determines whether oscillation is *realised*, by tuning the fixed point into the feasibility window. Within this window, increasing $w_{EE}$ does promote oscillation — it drives $w_{EE}F_E'$ above the trace threshold, consistent with Li et al.'s conclusion. But this promotion is bounded above: excessive $w_{EE}$ either pushes $u_E$ past $\theta_E$ into saturation (collapsing $F_E'$, causing the trace itself to fail), or, even before saturation, drives $w_{EE}F_E'$ above the algebraic ceiling from Equation 22, producing a Saddle. The

[7] Whether the 10 three-fixed-point oscillations represent true (supercritical Hopf) limit cycles coexisting with the stable attractor or transient spirals that eventually decay would require computing the first Lyapunov coefficient at the unstable fixed point, which is a direction left for future investigation.

[8] With correction term it's slightly different.

complete picture is therefore that $w_{EE}$ acts as a *tuning parameter* with a finite effective range, not a monotonic promoter of oscillation.

We refer to the following papers:

**Inhibitory self-feedback.** Li et al. (2022):

> The present results reveal that, on one hand, the inhibitory self-feedback loop is not conducive to the generation of gamma oscillations, and increased inhibitory self-feedback strength facilitates the enhancement of the oscillation frequency. On the other hand, the excitatory self-feedback loop promotes the generation of gamma oscillations [...] Inhibitory and excitatory self-feedback loops play a complementary role in generating and regulating the gamma oscillation in Wilson-Cowan model.

**Time constants.** Keeley et al. (2019):

> The models' flexibility to capture the broad range of gamma behavior depends directly on the timescales that represent recruitment of the excitatory and inhibitory firing rates. In particular, we find that weakly modulated gamma oscillations occur robustly when the recruitment timescale of inhibition is faster than that of excitation.

# 4 Limitations

Several limitations of the present analysis should be noted.

**Decoupled optimisation.** The structural condition is obtained by independently maximising $F_E'$ and minimising the cost over $F_I'$. In the actual system, both derivatives are jointly determined by a single fixed point on the nullcline intersection, and may not simultaneously attain their optimal values. This makes Equation 33 a potentially loose bound: the gap between structural necessity and actual oscillation is quantified by the three-level framework in sufficiency, but no closed-form sufficient condition was obtained (see the section 5).

**Finite numerical scope.** The Monte Carlo study ($N = 5000$) found no three-fixed-point system with two unstable equilibria (U2/S0/Sad1), but this is an empirical observation, not a topological proof. The Poincaré index theorem permits a limit cycle enclosing all three such fixed points (index sum $= +1$), so this configuration cannot be ruled out in other parameter regimes.

**Sigmoid form.** All results assume the specific sigmoid with the correction term $F(0) = 0$. The original Wilson-Cowan formulation includes a refractory term $(1 - r_E)$ which modifies the effective gain; our conditions do not directly apply to that variant.

**Single-population time constants.** The model assumes each population is characterised by a single time constant. Distributed delays or synaptic filtering (as in Keeley et al. (2019)) may shift the oscillatory boundaries quantitatively, though the qualitative structure of the trace-determinant framework is preserved.

# 5 Future work

The Monte Carlo validation identifies a sharp empirical boundary: Level 3 at a unique fixed point yields oscillation at $\approx 90\%$, whereas Level 3 at one of three fixed points yields only $\approx 6\%$. Detailed logging reveals that all oscillating three-fixed-point systems have configuration U1/S1/Sad1 (one unstable, one stable, one saddle); no U2/S0/Sad1 configuration was observed. This suggests that three-fixed-point oscillation in the sampled parameter range occurs through basin-restricted limit cycles around the unstable spiral, rather than global limit cycles enclosing all three fixed points.

A natural next step is to derive, from parameters alone, whether there exists an $I_E^{\text{ext}}$ that produces a unique unstable fixed point. The key geometric ingredient is the E-nullcline fold. The trace condition requires $w_{EE}F_E' > 1$, which at the sigmoid midpoint demands $w_{EE} > 4/a_E$. But this is precisely the condition for the E-nullcline to be non-monotone (S-shaped), guaranteeing multiple fixed points at some input levels. As $I_E^{\text{ext}}$ increases, the E-nullcline shifts upward until a saddle-node bifurcation annihilates two fixed points, leaving a unique equilibrium. If this surviving fixed point satisfies the Hopf condition, a limit cycle is guaranteed by the Poincaré-Bendixson theorem.

The question therefore reduces to whether the Hopf bifurcation occurs in the one-fixed-point region of the $(I_E^{\text{ext}}, w_{EE})$ parameter plane — a codimension-2 problem involving the interaction between saddle-node and Hopf bifurcation curves. This Bogdanov-Takens / saddle-node-Hopf analysis is a standard topic in bifurcation theory (see Izhikevich (2007), Ch. 6; Kuznetsov (2004), Ch. 8) but requires case-specific computation for the Wilson-Cowan sigmoid. Completing this analysis would close the gap between structural necessity (Equation 33) and a fully parametric sufficient condition, eliminating the need for fixed-point enumeration.

# 6 Key References


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